SURVEYING & LEVELLING · NOTE SL03

Chain Surveying — Stations, Survey Lines, Offsets, Obstacles and Plotting

Course topics: SWCE1.4 L4

Principle

Chain surveying is simple triangulation: measure the sides of triangles directly in the field and take **no angles**.

Suitable	Not suitable
Small areas; fairly level, open ground	Large areas; crowded areas with many details and obstructions

- Divide the area into **well-conditioned** triangles — as nearly equilateral as possible — so measuring and plotting errors stay small.
- ICAR eCourse: a triangle with an angle below 30° or above 120° is ill-conditioned; its apex is not sharp and plots badly.

Stations and lines

Item	What it is
Main stations	Control the boundaries; joined by the main survey (chain) lines; must be intervisible; chosen to avoid obstacles
Subsidiary (tie) stations	Points on the main lines from which you take offsets to details; symbol ⊙
Base line	The longest chain line; the framework of triangles is built on it; keep it on level ground as far as possible
Check (proof) line	Joins the apex of a triangle to a fixed point on the opposite side; its ground length must agree with its plotted length
Tie line	Joins two fixed points on main lines; checks accuracy and locates interior details such as paths and buildings

Field procedure

1. **Reconnaissance:** walk over the whole area, decide the best arrangement and draw an index sketch of boundaries, roads, buildings and major features on the first page of the field book.
2. **Mark the stations:** with wooden pegs if the survey runs beyond 1 day, or with poles if it finishes the same day. Leave the pegs until the survey is complete.
3. **Measure:** chain the lines and take the offsets; fix the direction of the base line with a prismatic compass; enter every reading in the field book.

ICAR eCourse kit for a chain survey: a 20 m metric chain, 10 arrows, a 15 m metallic tape, 3 ranging rods, an offset rod, a clinometer, a plumb bob, a cross staff or optical square, a prismatic compass with stand, 10 wooden pegs, a mallet, field books and a good pencil.

Offsets

- An offset is the perpendicular distance from a point on the chain line to an object.
- Use offsets to locate boundaries, buildings, roads, fences and wells.
- **Short offsets:** estimate the right angle by eye. **Long offsets:** set them with a cross staff or optical square.
- Keep offsets short, take one at every change of direction and record each in the field book.

Obstacles in chaining

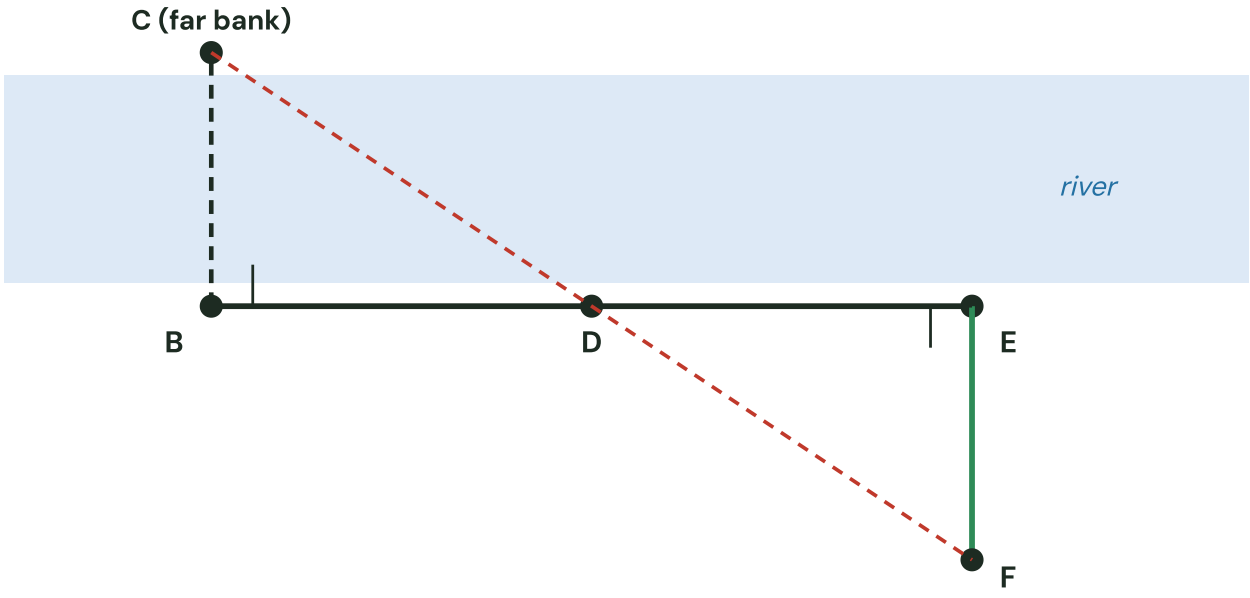
Typical obstacles are buildings, trees, walls, rivers and ponds.

Chaining past an obstacle

1. Set out 2 or 3 equal offsets with a cross staff or optical square.
2. Carry the chain line forward beyond the obstacle on these offsets.
3. Check that the poles on both sides stay in line.

Chaining across a river

1. Select B and C on opposite banks on the chain line.
2. Erect a perpendicular BE along the near bank and bisect it at D.
3. At E erect a perpendicular EF, and fix F where F, D and C fall in one line.
4. Measure EF: BC = EF.



Chaining across a river: equal triangles on either side of D

For a wide river, make DE = 1/3 BD; then BC = 3 × EF.

Survey notes, field book and map symbols

- **Survey notes** are the permanent record: title, location and purpose, names of staff, date and weather, numerical data and sketches.

- **Field book:** two central lines represent the chain line; enter measurements from the bottom of the page upwards; write offsets on the left or right; start each survey line on a fresh page.
- **Map symbols:** use standard cartographic symbols — road, well, fence, culvert, terrace, drain, building — so everyone reads the map the same way.

Plotting the survey

1. **Select the scale** to suit the purpose of the map, the area and the detail needed. Common engineer's scales: 1, 2, 3, 4, 5 or 6 m = 1 cm.
2. **Orient the plan:** plot it in its true position with respect to north, using the direction of the first chain line from the prismatic compass. Keep north at the top and leave a 1–2 cm margin on all sides.
3. **Plot the triangles:** draw the base line to scale (use a protractor if its bearing is known); with a beam compass, swing arcs of the other two sides — their intersection is the apex. Continue until the framework is complete.
4. **Plot the offsets:** draw perpendiculars, scale off the offset lengths and draw the features with conventional symbols.

A finished map shows: title, north arrow (true or magnetic), legend, graphical scale, project purpose, name of the tract, contour interval (if any), surveyor and draftsman, and date of survey.

Sources: ICAR eCourse 'Surveying and Leveling' — course content by Dr D. Tamil Mani (Dept. of Soil & Water Conservation Engg., AEC&RI, TNAU Coimbatore), reviewed by Dr V. M. Abdul Hakkim (KCAET, Kerala Agricultural University); published as an AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. In chain surveying, the area is divided into

(a) squares
(b) well-conditioned triangles
(c) circles
(d) trapeziums only

Q2. A triangle is ill-conditioned if any angle is

(a) between 30° and 120°
(b) less than 30° or more than 120°
(c) exactly 60°
(d) more than 90°

Q3. The longest chain line on which the framework is built is the

(a) tie line
(b) check line
(c) base line
(d) offset line

Q4. In the river-crossing method with BD = DE, the width BC equals

(a) DE
(b) EF
(c) 2 × EF
(d) EF/2

Q5. If DE is made 1/3 of BD, then BC equals

(a) EF/3
(b) EF
(c) 3 × EF
(d) 9 × EF

Q6. In a field book, chainages are entered

(a) from top to bottom
(b) from bottom to top
(c) left to right
(d) in any order

Answers

Q1 — (b) well-conditioned triangles

Q2 — (b) less than 30° or more than 120° — *ICAR eCourse definition.*

Q3 — (c) base line

Q4 — (b) EF — *Triangles BCD and EFD are equal.*

Q5 — (c) 3 × EF

Q6 — (b) from bottom to top
